SHARE Digital Twin: System Architecture & Dual-Stream Pipeline

Cardiovascular Autonomic Baseline Modeling & Exploratory Event Prediction | PhysioNet SHAREE Cohort (N=139, 17 Confirmed Vascular Events; 12.2% Prevalence)

16:9 Presentation Vector | PhysioNet ODC-By v1.0 | Research Prototype Only

Not for clinical diagnostic, urgent care, or treatment use

Files excluded from intended public repository by .gitignore

*.gitignore exclusion policy for repo distribution; does not guarantee privacy or imply audited publication history.

Stream 1: Static Historical / Clinical Data — 12 Features

Patient Baseline Intake & Echo Records (info.txt)

- **Demographics:** Age, Sex, Weight, Height, BSA, BMI, Smoker
- **Hemodynamics:** Systolic BP, Diastolic BP (mmHg)
- **Morphology:** IMT MAX, LVMI (g/m^2), Ejection Fraction (EF%)

Excluded Local File: share_static_cohort.csv
139 patient records × 12 clinical columns | Input to Static RF

Stream 2: Dynamic HRV Time-Series — 42 Features

24h Ambulatory Holter ECG (128 Hz, 3 Leads, QRS)

15-Minute Epoch Windowing (Non-Overlapping)
5 HRV Markers: HR (bpm), Mean RR (ms), SDNN, RMSSD, pNN50
→ 12,587 Windowed States across cohort (share_cohort_twin_states.csv)

24-Hour Summary Extraction (42 Dynamic Features):
• 8 stats per marker: Mean, Std, Min, Max, Median, First, Last, Change (5 × 8 = 40) + Valid Windows + Hours

Excluded Local Raw & Derived Files:
- share_data/*.dat, *.hea, *.qrs (~4.1 GB Holter waveforms)
- share_patient_level_dataset.csv (42 dynamic features)
- *oof_predictions.csv (Contains true patient event labels)
- *.joblib (Quarantined trained model checkpoints)

[Quarantined by .gitignore: Excluded from public repository]

Public Bundled Alternative: Synthetic Cohort — Public Git

Zero-Credential Evaluation Safeguard (demo_data/)
- Pre-generated demo patients: DEMO-001, DEMO-002, DEMO-003
- Generated via generate_synthetic_demo.py (Fixed Seed)
- Zero overlap with PhysioNet SHAREE records; all labels=0
- Runs out-of-the-box from a clean clone without real data

CLI Launch: python experiment_11_dashboard.py --demo

Stream 1: 12 Static Features Bus

Epochs

Twin State Engine (twin_state_engine.py) — Personalized

1. Individual Autonomic Baseline Calibration
- Calculates personalized median baseline across patient's 24h recording
- Avoids misclassifications from arbitrary population-level cutoffs

2. Outlier-Robust Scaling: Median Absolute Deviation (MAD)

```
Robust Z-Score_i = (x_i - Median) / (1.4826 × MAD)
```

Resistant to Holter recording noise artifacts & ectopic beats

3. Multi-Scale Twin State Deviation Metric

```
State Deviation = (1/5) × Σ |Robust Z-Score_i|
```

Computed across 5 markers: HR, Mean RR, SDNN, RMSSD, pNN50

4. Engineering Stability Bands (Non-Diagnostic):

Near Baseline	Moderate Shift	High Deviation
Deviation < 1.0	1.0 ≤ Dev < 2.0	Deviation ≥ 2.0

- Measures mathematical deviation from personal diurnal median.
- Does not denote clinical severity or immediate cardiovascular risk.

Stream 2: 42 Dynamic Features Bus

Perturbations

What-If Simulator (what_if_simulator.py) — Perturbation

Bounded Mathematical Perturbation Engine
- Evaluates hypothetical physiological state shifts interactively

Hard Parameter Validation Bounds:
- Heart Rate Adjustment: **±10% max** (physiological bound)
- SDNN / RMSSD / pNN50: **±20% max**
- Strictly immutable baseline: Patient median is never altered

Reciprocal Electrophysiological Coupling Formula:

```
Mean_RR_ms = 60,000 / HR_bpm
```

Preserves electrophysiological consistency upon heart rate shifts

Simulation Output & Strict Non-Causal Boundaries:
✓ Instantly recomputes robust z-scores & state deviation
✓ Visualizes counterfactual stability deltas on UI radar
✗ **Refuses fabricated clinical risk predictions or probabilities**
✗ **Non-causal: Does not simulate drug therapy or interventions**
(Isolated from ML models; verified by 10/10 automated tests)

Exploratory Machine Learning Pipelines

5-Fold Stratified Patient CV | Cohort N=139, 17 Events (12.2% Prevalence)

Dynamic Random Forest (Exp 11) — PRIMARY MODEL

Input: 42 Dynamic 24h Summary Features
500 trees | class_weight='balanced' | Stratified 5-Fold CV

VERIFIED OUT-OF-FOLD (OOF) BENCHMARK METRICS:
OOF ROC-AUC: 0.657 (0.6572) | OOF AP: 0.259 (0.2590)
Mean Fold ROC-AUC: 0.699 ± 0.163 | AP vs. baseline prevalence: 0.259 vs 0.122

Primary Pipeline Flow → P_dynamic (Default Output)
Pure 100% dynamic signal | Zero static weighting applied

P_dyn

PRIMARY

P_stat

Static Clinical RF (Exp 13 Model) — 12 Features

Demographics, Blood Pressure, LVMI, Ejection Fraction
OOF ROC-AUC: 0.526 | OOF AP: 0.137 | Mean Fold: 0.561 ± 0.152
Static markers alone exhibit near-chance discriminative power
Feeds exclusively into the Optional Late Fusion Box below

P_fus 80/20

Early Fusion Concatenation Comparison (54 features):
Concatenated RF OOF ROC-AUC: 0.647 | OOF AP: 0.232 (Underperforms dynamic-only)

Optional Late Fusion Box (Dual-Stream Blend) — OPTIONAL MODE

80% Dynamic + 20% Static Prior Weighting Formula:

```
P_fusion = 0.80 × P_dynamic + 0.20 × P_static
```

Pre-specified empirical prior blend (Experiment 13)

EVALUATION BENCHMARK METRICS (5-FOLD STRATIFIED CV):
OOF ROC-AUC: 0.655 (0.6553) | OOF AP: 0.252 (0.2522)
Mean Fold ROC-AUC: 0.689 ± 0.165 | AP vs baseline: 0.252 vs 0.122

- **Why 80/20?** Empirical engineering prior acknowledging that dynamic HRV carries substantially higher predictive signal (AUC 0.657) than static clinical markers alone (AUC 0.526).
- **Non-Validation Notice: Weighting is an investigational heuristic** and is NOT clinically validated, guideline-endorsed, or FDA-cleared.
- **Only activated when query ?mode=dual_stream is requested.**
- **Default predictions use NO static weighting (P_dynamic pure).**

Outputs P_fusion → Activates Dual-Stream mode on Clinician Dashboard

Public Demo Cohort Bus (--demo)

P_dynamic — Default prediction — DEFAULT OUTPUT

Single-Stream Dynamic Random Forest Inference
- Derived purely from 42 dynamic HRV summary features
- **Zero static weighting applied** (No 80% weight implied)
- Verified Benchmark Performance: **OOF ROC-AUC 0.657**
- **Default API contract: /api/patient/<id> (active_mode: "dynamic")**

P_fusion — Optional Dual-Stream Prediction — OPTIONAL

- Output of Late Fusion Box: 0.80 × P_dynamic + 0.20 × P_static
- Activated exclusively via query: `?mode=dual_stream`
- Verified OOF ROC-AUC: 0.655 | Displays both stream contributions

Clinician Dashboard

experiment_11_dashboard.py | Flask REST API & Web UI — Flask :5000

DUAL EXECUTION ARCHITECTURE:
- Live Research Mode: python experiment_11_dashboard.py
- Zero-Credential Demo: python experiment_11_dashboard.py --demo

Interactive User Interface Tabs (templates/dashboard.html):

Tab 1: Patient Overview & Static Profile
Baseline demographics, blood pressure, BMI, LVMI, and ejection fraction

Tab 2: Twin State & Baseline Radar
MAD-scaled robust z-score radar chart + multi-scale deviation meter

Tab 3: 24h Trajectory Timeline
Time-series of 15-min windows, diurnal trends & anomalous epoch alerts

Tab 4: Model Explanation & Feature Importances
Gini & permutation rankings (SDNN change, RMSSD max, Mean RR)

Tab 5: Counterfactual What-If Simulator
Interactive perturbation sliders (HR, SDNN, RMSSD) with live recomputation

REST API CONTRACTS & ENDPOINTS:

```
GET /api/patient/<id> → Default P_dynamic
GET /api/patient/<id>?mode=dual_stream → P_fusion (80/20)
POST /api/simulate → Recomputed Z-scores
```

MANDATORY UI ETHICAL & SCIENTIFIC DISCLAIMERS:
- Prominently surfaces 17/139 event imbalance warning across tabs
- Explains that Twin Deviation is an engineering stability score
- Discloses absence of external cohort validation
- In demo mode without .joblib: displays labeled placeholder UI scores
- **Explicitly disclaims medical device status or clinical diagnostic use**

SYSTEM LEGEND: Excluded by .gitignore | Dynamic Stream (Exp 11) | Static Stream (Exp 13) | Optional Late Fusion (80/20) | Twin Engine & Simulator | Public Demo Bundle | Default Output Flow

PhysioNet SHAREE Database DOI: 10.13026/C2RC7C | Melillo et al. PLoS ONE (2015) | Open Data Commons Attribution License v1.0 (ODC-By v1.0).

Architecture specification for Happiest Health Digital Twin Challenge 2026. Research prototype only — not validated for clinical care.